

Indian Institute of Technology Madras

DA5400W - Foundations of machine learning

Assignment on Classification

Due date: April 17, 2026, 11:59 pm

Instruction

1. Assignment shall be submitted on the due date. Late submissions will not be entertained.
2. Students must submit only their own work. Copying or sharing assignments will lead to serious academic penalties for everyone concerned.
3. Feel free to use python code wherever you feel necessary and only provide the main results in the submission, unless advised otherwise.

Problems

1. **(Solve the problem by hand using a calculator)** Consider the following dataset used to classify industrial components as **Pass** or **Fail**:

Sample	Temperature (°C)	Vibration (mm/s)	Class
1	40	1.0	Pass
2	42	1.2	Pass
3	45	1.1	Pass
4	60	4.5	Fail
5	62	4.8	Fail
6	65	5.0	Fail

A new sample has features (50, 1.5).

- (a) Classify the new sample using the 3-nearest neighbors (3-NN) classifier with Euclidean distance, **without** performing any feature scaling.
 - (b) Standardize each feature using the mean and standard deviation computed from the training data, and then classify the new sample again using the 3-NN classifier with Euclidean distance.
 - (c) Based on your results, comment on the importance of feature scaling in the KNN classifier.
2. A training dataset and a test dataset for a binary classification problem are provided in the accompanying Excel files. The target variable is **Class**, which takes values **Normal** and **Fault**. The training dataset is imbalanced, with the minority class corresponding to **Fault**.

Use the given datasets to answer the following questions.

- (a) Train a K-nearest neighbors (KNN) classifier on the training dataset using all the input features. Before training, apply appropriate feature scaling. Use Euclidean distance and choose a suitable value of k using a reasonable validation strategy.
- (b) Using the model obtained in part (b), predict the class labels for the test dataset.
- (c) Based on the predictions from part (c), compute the following performance measures on the test dataset:

- (i) confusion matrix,
 - (ii) precision,
 - (iii) recall,
 - (iv) F1-score,
 - (v) balanced accuracy.
- (d) Since the training dataset is imbalanced, apply **two different methods** to handle class imbalance and retrain the KNN classifier. You may choose any two of the following:
- random oversampling of the minority class,
 - random undersampling of the majority class,
 - synthetic oversampling such as SMOTE,
 - distance-weighted KNN.
- (e) For each of the two imbalance-handling methods chosen in part (d), retrain the classifier and evaluate it on the same test dataset. Report:
- (i) the chosen method,
 - (ii) the value of k used,
 - (iii) the confusion matrix,
 - (iv) the F1-score,
 - (v) the balanced accuracy.
- (f) Compare the results of the baseline KNN classifier in part (c) with the two imbalance-handling approaches in part (e). Comment on whether handling class imbalance improves performance, especially for the minority class.
3. Consider the following dataset of the students of class. Apply the Naive Bayes classifier on the dataset and estimate the probability of a student excelling in the end-semester exam given her mid-semester performance is “Average” and has submitted all her assignments. Please ensure you show all the necessary calculations by hand.

Mid Sem Performance	Submitted Assignments	Aced End Sem
Below Average	False	False
Below Average	False	False
Below Average	True	True
Below Average	False	False
Average	True	False
Average	True	True
Average	False	False
Average	True	True
Average	True	True
Above Average	False	True
Above Average	True	True

Table 1: Data set for Q10

4. Consider a Multinomial Naive Bayes classifier for a classification problem with K classes and a vocabulary of size V . Let θ_{jc} denote the probability of observing feature (or word) j in class c , where

$$\sum_{j=1}^V \theta_{jc} = 1, \quad \theta_{jc} \geq 0.$$

Suppose that for class c , the total count of feature j aggregated over all training samples belonging to class c is N_{jc} .

- (a) Write down the likelihood function of the training data for class c under the multinomial model.
- (b) By taking the logarithm of the likelihood and using the constraint

$$\sum_{j=1}^V \theta_{jc} = 1,$$

derive the maximum likelihood estimator (MLE) of θ_{jc} .

- (c) State the final MLE expression clearly in terms of the counts N_{jc} .

5. Show that $\frac{\partial \sigma}{\partial a} = \sigma(1 - \sigma)$ for the logistic sigmoid function.
6. A multinomial logistic regression classifier is used to classify a production batch into one of three categories:

$$C_1 = \text{Accept}, \quad C_2 = \text{Rework}, \quad C_3 = \text{Reject}.$$

For a given batch, the model produces the following class scores:

$$z_1 = 2.4, \quad z_2 = 1.8, \quad z_3 = 0.6.$$

- (a) Compute the softmax probabilities for all three classes and mention the predicted class.
- (b) Compute the odds ratio

$$\frac{p(C_1 | x)}{p(C_2 | x)}$$

and interpret its meaning.

- (d) Suppose that due to a calibration shift, the same constant 5 is added to all three scores, would the classifier predictions be affected.
 - (e) Suppose instead that only the score of class C_3 increases from 0.6 to 2.0, while the other two scores remain unchanged. Recompute the softmax probabilities and comment on how the prediction confidence changes.
7. Consider the plotted data points in Fig. 1. The problem involves two features (X1, and X2) and two classes (0, 1). There are two unknown points in the figure. Answer the following questions:
 - (a) Develop a decision tree-based classifier with the available data in the figure (Class 0 and Class 1), with the Unknown data points being the test data.
 - (b) Provide calculations for determining each split, including root nodes. One can use misclassification rate, entropy, and Gini as criteria.
 - (c) If you include one of the unknown points and arbitrarily assign one of the classes, will it change the decision tree?
 8. Train a decision tree using the first 16 observations. The remaining two observations can be used for the test data. Then, answer the following questions regarding the fitted decision tree
 - (a) Is it possible to prune the tree? Which criteria can be used to prune the tree?
 - (b) Any feature can be dropped? If "yes", which feature should be dropped?
 - (c) Is it possible to get pure labels for the fitted tree?

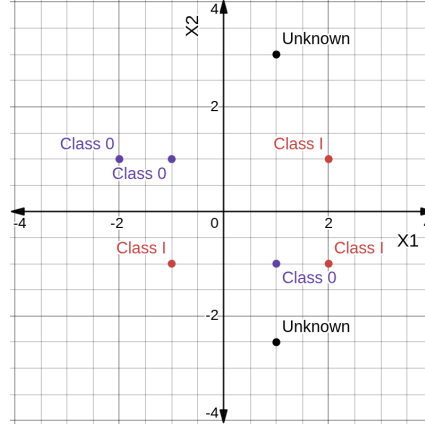


Figure 1: Figure For Question 3

Table 2: Dataset for Question 4

Sr. No	Income	Application Usage	Age Group	Favorite Color	Classes
1	High	Daily	>25	Red	Yes
2	High	Daily	>25	Blue	Yes
3	High	Daily	>25	Green	Yes
4	High	Daily	>25	Red	Yes
5	High	Daily	>25	Blue	Yes
6	High	Daily	>25	Green	Yes
7	High	Daily	>25	Red	Yes
8	High	Daily	>25	Blue	Yes
9	High	Daily	>25	Green	Yes
10	High	Daily	>25	Red	No
11	Low	Monthly	<25	Blue	No
12	Low	Weekly	>25	Green	No
13	Low	Daily	25-45	Red	No
14	Low	Monthly	>25	Blue	No
15	Low	Weekly	<25	Green	No
16	Medium	Weekly	<25	Red	Yes
17	Medium	Weekly	<25	Red	Yes
18	Medium	Weekly	<25	Blue	No

9. Consider a data set comprising 400 data points from class C1 and 400 data points from class C2. Suppose that a tree model A splits these into (300, 100) at the first leaf node and (100, 300) at the second leaf node, where (n, m) denotes that n points are assigned to C1 and m points are assigned to C2. Similarly, suppose that a second tree model B splits them into (200, 400) and (200, 0) at the first leaf and the second leaf nodes.

- Evaluate the misclassification rates for the two trees and show that they are equal.
- Evaluate the cross-entropy and Gini index for the two trees and show that they are both lower for tree B than for tree A.

10. Consider the following binary classification dataset with labels $y_i \in \{-1, +1\}$:

Sample i	True label y_i	Prediction of weak classifier $h_1(x_i)$
1	+1	+1
2	+1	+1
3	+1	-1
4	-1	-1
5	-1	+1
6	-1	-1

Assume that initially all training samples have equal weights:

$$w_i^{(1)} = \frac{1}{6}, \quad i = 1, \dots, 6.$$

AdaBoost is applied using the above weak classifier as the classifier obtained in the first round.

- Compute the weighted classification error ϵ_1 of the weak classifier h_1 .
- Compute the classifier weight
- Update the sample weights and normalise them to 1. Report the updated weights for all six samples.
- Based on the updated weights, identify which samples will receive greater emphasis in the next boosting round. Briefly explain why.
- Suppose that in the second round, the weak classifier h_2 makes mistakes only on samples 2 and 4. Using the updated weights from part (c), compute the weighted error ϵ_2 of h_2 .
- Without carrying out the full weight update again, explain whether h_2 is expected to receive a larger or smaller classifier weight than h_1 , based on the value of ϵ_2 .